

Session 1 — Student Reading

From Circuits to Graphs, From Graphs to Space

LaPSEE Minicourse
Introduction to Power Networks as Intelligent Graphs

Abstract

This reading accompanies Session 1 of the LaPSEE minicourse. It introduces the graph representation of power networks, the classical topological metrics used for vulnerability assessment, and the bridge from graph adjacency to the spatial weight matrix $\mathbf{W}$ that underpins spatial econometrics. The worked example uses the IEEE 30-bus test system; the companion notebook `s01_graphs-and-space.ipynb` implements every concept in Python.

Learning objectives

By the end of this reading you will be able to:

- Build a graph representation of a power system and compute its degree, betweenness, closeness, eigenvector, and clustering-coefficient centralities.
- Construct a spatial weight matrix $\mathbf{W}$ from an adjacency matrix and from geographic coordinates (contiguity, inverse distance, k -nearest neighbours).
- Compute and test Moran's I statistic on a vector of nodal observations and interpret a Moran scatterplot and a LISA cluster map.
- Recognise the three canonical spatial regression models (SAR, SEM, SDM) and name the right diagnostic for choosing among them.

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1 Motivation

Power system analysis traditionally operates on two mathematical objects: the bus admittance matrix $\mathbf{Y}_{\text{bus}}$ and the flow equations of the network. These objects give us *local* answers — voltages, angles, flows, losses. They say very little about how a given bus is embedded in the larger network, or about how observations at neighbouring buses relate to each other statistically.

Complex network theory and spatial econometrics fill those blind spots.

Key idea

The first-order contiguity spatial weight matrix of a power grid *is* the adjacency matrix of its graph representation. Graph theory and spatial econometrics are two analytical vocabularies speaking about the same combinatorial object.

Throughout the course the narrative moves from *structure* (Session 1) to *space* (Session 2) to *behaviour* (Session 3). This session establishes the dual foundation that the rest of the course uses.

2 Graph model of the power network

Definition 2.1. A power system with n buses and m transmission elements is modelled by the undirected graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$, where $\mathcal{V} = \{1, \dots, n\}$ is the set of buses and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of branches (lines plus transformers). We allow weighted edges $w : \mathcal{E} \rightarrow \mathbb{R}_{>0}$ to encode electrical coupling.

The *adjacency matrix* $\mathbf{A} \in \{0, 1\}^{n \times n}$ is defined by $\mathbf{A}_{ij} = 1$ if $(i, j) \in \mathcal{E}$ and zero otherwise. Three consequences are worth internalising:

1. $\mathbf{A}$ is symmetric for undirected graphs; $\mathbf{A} = \mathbf{A}^\top$.
2. $d_i := \sum_j \mathbf{A}_{ij}$ is the degree of bus i .
3. $(\mathbf{A}^k)_{ij}$ counts walks of length k from i to j ; this is the algebraic bridge to centrality measures that depend on long-range paths.

2.1 Weighted adjacency

Purely topological analysis discards information about how strongly two buses are coupled. Three common weighting schemes are:

- **Impedance-weighted:** $w_{ij} = 1/|z_{ij}|$ or, in DC studies, $1/x_{ij}$. Reflects electrical proximity.
- **Flow-weighted:** $w_{ij} = |P_{ij}^{\text{flow}}|$ from a DC power flow or from historical SCADA data. Reflects operational reality.
- **Distance-weighted:** $w_{ij} = 1/d(i, j)$ if geographic coordinates are available. Reflects physical span.

The *admittance matrix* $\mathbf{Y}_{\text{bus}}$ is itself a weighted adjacency up to a sign convention: its off-diagonal non-zeros mirror $\mathbf{A}$, and for the linearised DC model the imaginary part $\mathbf{B} = \Im(\mathbf{Y}_{\text{bus}})$ is a weighted graph Laplacian.

Remark. Unless the research question requires direction (e.g. market signals or control hierarchies), prefer undirected models for structural analysis. Directionality can always be reintroduced later as an edge attribute.

3 Topological centrality metrics

Different centrality measures encode different hypotheses about which bus matters. Five are standard in power-systems applications.

3.1 Degree centrality

$$C_D(i) = \frac{d_i}{n-1}.$$

Simple, local, cheap. It captures only immediate connectivity and is often nearly flat on real grids. Rarely sufficient on its own.

3.2 Betweenness centrality

$$C_B(i) = \sum_{s \neq t \neq i} \frac{\sigma_{st}(i)}{\sigma_{st}},$$

where σ_{st} counts shortest s - t paths and $\sigma_{st}(i)$ those passing through i . High C_B indicates a *traffic* role: the bus lies on many shortest routes. In power systems, the concept extends naturally to *electrical betweenness*, where paths are weighted by line admittance or by the current injected by unit fictitious transactions (Wang et al. 2011).

3.3 Closeness centrality

$$C_C(i) = \frac{n-1}{\sum_{j \neq i} d(i, j)},$$

using graph-distance $d(i, j)$ (hop count or weighted shortest path). A node with high closeness is close *on average* to every other node, so a disturbance at i propagates quickly through the network.

3.4 Eigenvector centrality

Let $\mathbf{A}\mathbf{v} = \lambda_{\max}\mathbf{v}$ with $\|\mathbf{v}\|_2 = 1$ and $\mathbf{v} \geq 0$ (Perron–Frobenius). Then

$$C_E(i) = v_i = \frac{1}{\lambda_{\max}} \sum_j \mathbf{A}_{ij} v_j.$$

A bus is influential if its neighbours are influential — a recursive, global measure. PageRank is a damped variant used when influence is partly local.

3.5 Clustering coefficient

$$C_{cl}(i) = \frac{2t_i}{d_i(d_i - 1)}, \quad t_i = \text{triangles through } i.$$

Measures how meshed the neighbourhood of i is. High C_{cl} implies redundancy (more alternative paths); low C_{cl} implies tree-like vulnerability.

3.6 From centrality to vulnerability

Classical N-1 analysis ranks elements by the impact of their outage. Topological centralities give a *cheap prior* on that ranking. Two practical guidelines:

1. Use multiple centralities simultaneously. Betweenness and eigenvector rarely agree, and disagreement is itself informative.
2. Cross-check with physical simulation. Flow-betweenness correlates best with cascading-failure extent (Albert et al. 2004; Cuadra et al. 2015), but *correlates* does not mean *replaces*.

Pitfall

A high centrality score is a *hypothesis*, not a conclusion. Always validate the top-k candidates against an N-1 or cascading-failure study before recommending an investment or a protection scheme.

4 Worked example on the IEEE 30-bus system

The IEEE 30-bus test system has $n = 30$, $m = 41$ branches, and six generators. A representative set of metrics, rounded to four significant figures, appears in Table 1. All numerical values are produced by the companion notebook; you should be able to reproduce them with a fresh kernel.

Table 1: Top five buses by each centrality on IEEE 30 (reproducible from the notebook).

Rank	Degree	Betweenness	Closeness	Eigenvector	Clustering
1	6	6	6	6	10
2	10	10	10	10	22
3	12	12	9	9	12
4	9	9	12	4	6
5	4	28	4	2	9

Bus 6 is a strong hub under every measure except clustering. That consistency is a signal: Bus 6 connects diverse sub-areas, making it structurally critical. Bus 28 appears only in the betweenness ranking — it sits on a bottleneck corridor but is otherwise poorly connected. Bus 22 has high clustering because it is embedded in a dense mesh, which means *redundancy*, not criticality.

5 From adjacency to spatial weights

Spatial econometrics studies regression models in which observations at different locations are dependent. Dependence is encoded by a *spatial weight matrix* $\mathbf{W} \in \mathbb{R}^{n \times n}$ with $\mathbf{W}_{ii} = 0$ and $\mathbf{W}_{ij} \geq 0$.

5.1 Common constructions

- **First-order contiguity:** $\mathbf{W} = \mathbf{A}$. Two buses are neighbours iff they share a branch. This is the *native* power-system choice.
- **k -nearest neighbours:** pick the k closest buses by graph or geographic distance. Guarantees no islands but breaks symmetry.
- **Inverse distance:** $\mathbf{W}_{ij} = 1/d(i, j)^\alpha$ with a cutoff $d_{\max}$. Requires geographic coordinates.
- **Higher-order contiguity:** $\mathbf{W}_{ij}^{(k)} = 1$ iff the shortest-path length between i and j is exactly k . Useful to probe spillover range.

5.2 Row standardisation

Define the row-standardised matrix

$$\tilde{\mathbf{W}}_{ij} = \frac{\mathbf{W}_{ij}}{\sum_k \mathbf{W}_{ik}}, \quad \sum_j \tilde{\mathbf{W}}_{ij} = 1.$$

Row standardisation has two virtues: it makes the product $\tilde{\mathbf{W}}\mathbf{y}$ the vector of neighbourhood averages, and it is required for most spatial estimators.

Pitfall

If $\mathbf{W}$ has isolated rows (buses without neighbours), row standardisation divides by zero. In power-grid graphs this happens when the contiguity scheme is too restrictive, e.g. second-order only. Always check $\min_i \sum_j \mathbf{W}_{ij} > 0$ before standardising.

6 Global and local spatial autocorrelation

6.1 Moran's I

Given $\mathbf{y} \in \mathbb{R}^n$ with mean $\bar{y}$, Moran's I is

$$I = \frac{n}{S_0} \frac{\sum_{i,j} \mathbf{W}_{ij} (y_i - \bar{y})(y_j - \bar{y})}{\sum_i (y_i - \bar{y})^2}, \quad S_0 = \sum_{i,j} \mathbf{W}_{ij}.$$

It is the correlation between y_i and the weighted average of its neighbours' values. Under the null of no spatial structure, $\mathbb{E}[I] = -1/(n-1)$, which converges to zero as $n \rightarrow \infty$. Positive values indicate spatial clustering; negative values, a checkerboard-like pattern.

Significance is assessed by a permutation test: shuffle $\mathbf{y}$ across buses N_p times and compare the observed I to the empirical distribution.

6.2 Local Moran (LISA)

Global I hides heterogeneity. The local indicator

$$I_i = \frac{y_i - \bar{y}}{m_2} \sum_j \mathbf{W}_{ij} (y_j - \bar{y}), \quad m_2 = \frac{1}{n} \sum_k (y_k - \bar{y})^2,$$

decomposes the global value into per-bus contributions and classifies each bus into one of four quadrants of the Moran scatterplot:

- High-High and Low-Low: *spatial clusters* of similar values.
- High-Low and Low-High: *spatial outliers*.

LISA is the workhorse of hotspot detection (Anselin 1995; Anselin and Rey 2014).

7 Spatial regression models

Three specifications dominate cross-sectional spatial analysis.

$$\text{SAR: } \mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \quad (1)$$

$$\text{SEM: } \mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \mathbf{u}, \quad \mathbf{u} = \lambda \mathbf{W}\mathbf{u} + \boldsymbol{\varepsilon}, \quad (2)$$

$$\text{SDM: } \mathbf{y} = \rho \mathbf{W}\mathbf{y} + \mathbf{X}\boldsymbol{\beta} + \mathbf{W}\mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon}. \quad (3)$$

7.1 Reduced form of SAR

Rearranging (1),

$$\mathbf{y} = (\mathbf{I} - \rho \mathbf{W})^{-1}(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}) = \left(\sum_{k \geq 0} (\rho \mathbf{W})^k \right) (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}).$$

A shock at bus j reaches bus i with multiplier $((\mathbf{I} - \rho \mathbf{W})^{-1})_{ij}$, composing effects from first-order neighbours, second-order neighbours, and so on. Stability requires $|\rho| < 1/|\mu_{\max}(\mathbf{W})|$, which equals 1 for a row-standardised $\mathbf{W}$.

7.2 Direct, indirect, and total effects

For any regressor x_r ,

$$\frac{\partial \mathbf{y}}{\partial x_r} = (\mathbf{I} - \rho \mathbf{W})^{-1} \boldsymbol{\beta}_r.$$

The *direct effect* is the average of diagonal entries; the *indirect (spillover) effect* is the average of off-diagonal entries; the *total effect* is the row-average of the matrix. LeSage and Pace's decomposition is the only legitimate way to interpret SAR and SDM coefficients: the raw $\boldsymbol{\beta}$ does *not* correspond to a marginal effect (LeSage and Pace 2009).

7.3 Choosing between SAR, SEM, SDM

A practical decision workflow:

1. Estimate OLS; test residuals for spatial autocorrelation (LM-lag and LM-error).
2. If LM-lag is significant but LM-error is not: lean SAR.
3. If the reverse: lean SEM.
4. If both are significant, or if theory predicts spillovers, estimate SDM and test it against SAR (LR test on $\boldsymbol{\theta} = 0$).

LeSage's influential recommendation is that **SDM is the safe default** for cross-sectional spatial analysis: it nests SAR and OLS and the cost of including $\mathbf{W}\mathbf{X}$ is small compared to the risk of biased interpretation.

8 Bridge to the notebook

The companion notebook `s01_graphs-and-space.ipynb` is split into two labs that mirror the syllabus time table.

Lab 1a (20 minutes). Implement everything from Sections 2–4 on the IEEE 30-bus system: build the graph with `networkx` and `pandapower`, compute all five centrality measures, and visualise critical nodes.

Lab 1b (10 minutes). Implement Sections 5–6: build a `libpysal` weight object from the adjacency, compute global Moran’s I on nodal load, draw the Moran scatterplot, and map LISA clusters.

Each function or cell in the notebook has a header comment pointing back to the equation number in this reading. Use the two documents in parallel.

9 Exercises

The following exercises are not graded but are expected work for this session. Full solutions appear in the *professor reading* that accompanies this file; please attempt them before looking up the solutions.

Exercise 9.1. Prove that the number of walks of length k from bus i to bus j in an undirected graph equals $(\mathbf{A}^k)_{ij}$. Use induction on k .

Exercise 9.2. Show that for a connected graph with n nodes the leading eigenvalue of $\mathbf{A}$ satisfies $\lambda_{\max} \geq \bar{d} = \frac{1}{n} \sum_i d_i$. In one sentence, what does equality mean?

Exercise 9.3. On the IEEE 30-bus case, compute the rank correlation (Spearman) between degree centrality and betweenness centrality. Report the value and interpret it in terms of how much information betweenness adds over degree.

Exercise 9.4. Construct the second-order contiguity matrix $\mathbf{W}^{(2)}$ and verify that its row sums are not constant. Identify one row with zero sum, if any, and explain what bus it corresponds to in the IEEE 30 topology.

Exercise 9.5. Run Moran’s I twice on nodal load: once with the contiguity $\mathbf{W}$ and once with a k -nearest neighbours $\mathbf{W}$ (pick $k = 4$). Explain why the values differ, and which one is more appropriate for analysing congestion.

Exercise 9.6 (Synthesis). Show that the SAR reduced form can be written as a Neumann-series expansion $\mathbf{y} = \sum_{k \geq 0} (\rho \mathbf{W})^k (\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon})$. Use this expansion to argue that betweenness centrality correlates with the *indirect effect* of a local shock. What assumption on $\mathbf{W}$ is needed for the series to converge?

10 Further reading

- Giuliano Andrea Pagani and Marco Aiello (2013). “The Power Grid as a Complex Network: A Survey”. In: *Physica A: Statistical Mechanics and its Applications* 392.11, pp. 2688–2700
- Zhifang Wang et al. (2011). “Electrical Betweenness Approach for Vulnerability Assessment of Power Grids”. In: *Physica A: Statistical Mechanics and its Applications* 390.23, pp. 4692–4701
- Lucas Cuadra et al. (2015). “A Critical Review of Robustness in Power Grids Using Complex Networks Concepts”. In: *Energies* 8.9, pp. 9211–9265
- Luc Anselin and Sergio J. Rey (2014). *Modern Spatial Econometrics in Practice: A Guide to GeoDa, GeoDaSpace, and PySAL*. GeoDa Press
- James LeSage and R. Kelley Pace (2009). *Introduction to Spatial Econometrics*. CRC Press
- Sergio J. Rey et al. (2023). *Geographic Data Science with Python*. CRC Press

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